Lab 6
Part 2 Question 6: Reflection

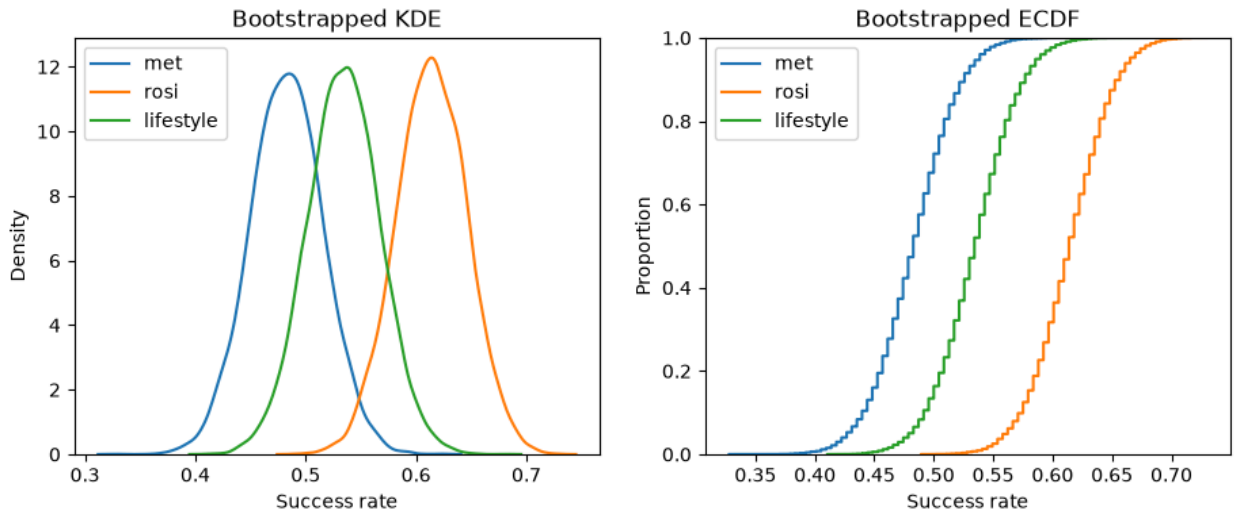
This evidence follows the data from a clinical trial considering three treatments for pre-diabetics in order to reduce the progression to full diabetes. The treatments that are focused on this clinical trial are Metformin (met), Rosiglitazone (rosi), and Lifestyle adjustments such as exercising or dieting. The results will demonstrate whether the treatments provide any sort of effectiveness for pre-diabetes.

The outcome of this dataset is binary, success or failure. Because the participants were randomly assigned, the difference in success rates between arms can be treated as the average treatment effect. Through this trial, we observed that the Rosiglitazone treatment is predicted to be the most effective to slow the progression of the pre-diabetes. When crosstabulating the data, Rosiglitazone had a success rate of 61%, whereas the lifestyle change treatment had a 53% success rate, and the Metformin treatment had a 48% success rate. These rates can be seen in the table below. The percentage of success can be a bit biased since the sample size of each varies from one another.

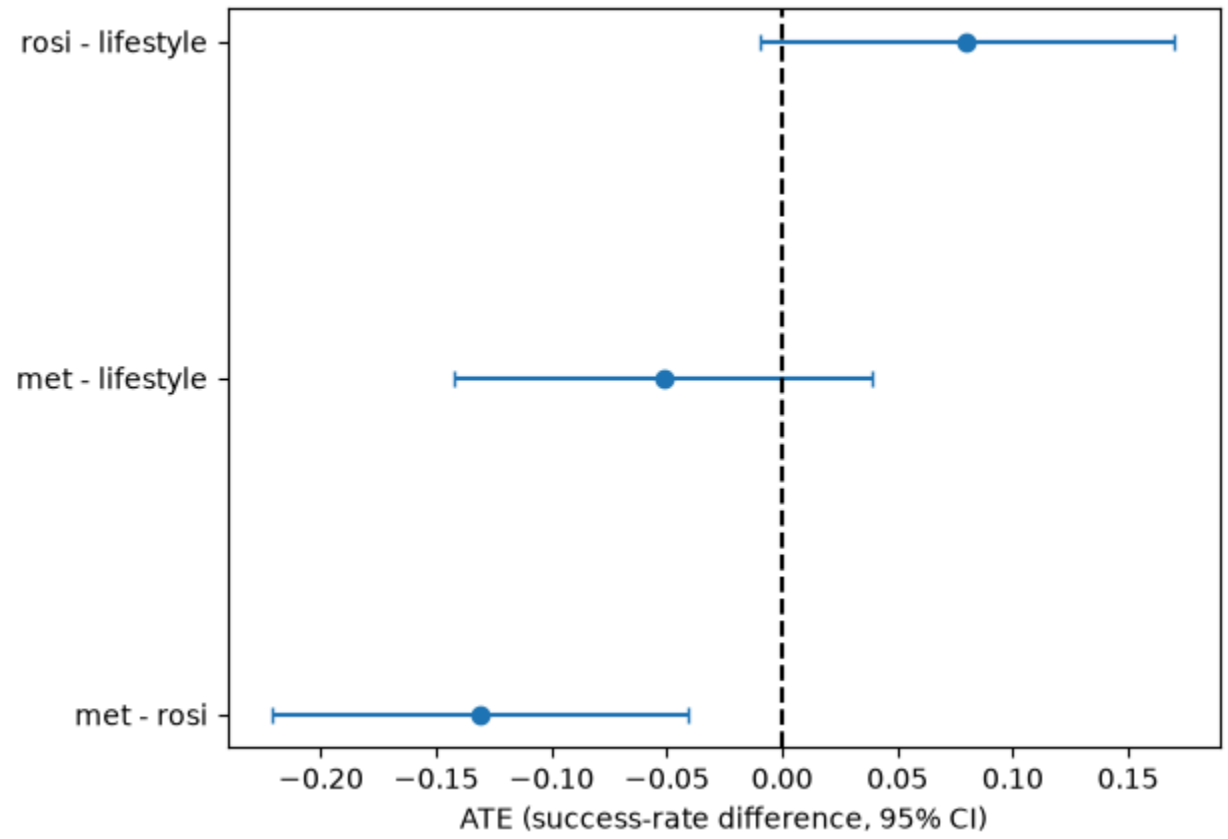
outcome	failure	success
treatment		
lifestyle	109	125
met	120	112
rosi	90	143

met to rosi: -0.131
met to lifestyle: -0.051
rosi to lifestyle: 0.08

We then repeated the same clinical trial over 10,000 times to see if the estimated success rate would vary for each treatment group if tested over and over again. This bootstrapping allows us to not rely on one set of results and to test if the best treatment holds consistent in the long run. After bootstrapping, we calculated the ECDF and KDE, which both showed that Rosiglitazone had the highest survival rate compared to Metformin and Lifestyle treatments.



From the ECDF graph, we can see that the Rosiglitazone survival rate is higher. We know this because the graph rises later than the Met and Lifestyle treatments. This is confirmed by the KDE graphs. The Rosiglitazone survival rate is higher than Met and Lifestyle, since the normal distribution is towards the right, meaning it has a higher success rate.



When looking at confidence intervals to assess if there was a statistically significant difference in the treatments, we once again see the Rosiglitazone treatment has better results. When comparing Rosiglitazone with Metformin, the confidence interval was negative on both sides, showing evidence of a significant difference. The Metformin and Lifestyle comparison was not significant, nor was the difference between Rosiglitazone and Lifestyle. From this analysis, we would suggest adopting the Rosiglitazone intervention since it continuously had the highest observed success rate, and performed significantly better than the Metformin intervention in this trial. However, we can't say whether the Rosiglitazone intervention is more effective than the lifestyle intervention, and we can't say if Lifestyle is better than Metformin. In these cases, we need more data to look further into significant differences of success between these interventions.